

Capturing traffic over Wi-Fi

Side Capture

Phone not able to import media

Privacy Analysis of Augmented Reality Glasses

1 Introduction

This experiment captures network traffic over Wi-Fi during the interaction between an Android phone and the Ray Ban Meta glasses, more specifically while executing touch commands. There was an unexpected issue during capture where the phone was not able to import the video recording from the glasses and was therefore deemed unsuccessful. However, I continued capturing network traffic whilst troubleshooting to determine if any relevant data could be observed during the failure. The issue was resolved in the end by rebooting the glasses. The hope (goal) for keeping this data collection is to have a view of the glasses's behavior under error conditions, which providing insights into how the device handles errors and reboots.

Captures: 3 different sets of traffic captures. The first captures the network traffic while the touch commands are being executed. The second captures the period after the import failure occurred and the other methods I tried to see if the glasses would reconnect after some time. The third captures the reboot process as well as an update that happened to occur right after.

Voice recording: Unfortunately the voice recording does not extend to the 3rd capture where the reboot took place.

2 Metadata

Time of experiment: Oct 14th 19:30

Duration of experiment: Roughly 30 minutes

IP addresses:

- Laptop - 192.168.0.215
- Redmi phone - 192.168.1.141
- Wi-Fi Network - 192.168.0.1
- ALFA Hotspot - 192.168.1.100/24
- Meta CDN (content delivery network) - 157.240.17.17

3 Touch Commands used

- Keep idle for 2 minutes
- Power On
- Pairing device
- Open Meta View
- Short click capture button to take a photo
- Upload image to phone
- Long click capture button to take a video recording

- Short click capture button to stop video recording
- Upload video to phone

4 Time Stamps

For capture 3, the following is some timelines that should be noted. The number is the packet number on Wireshark:

After 5462 - reboot phone

7230 - open meta app

8900 - import media

12500 - media done importing

13310 - start update

16415 - update complete and glasses restarted

5 Note